SSH Keyless Login & SCP Transfer Guide

A Practical Reference for Key Authentication and Remote File Operations (Windows to Linux)

PART 1 SSH Keyless Login Setup (Step-by-Step)

1 Step 1: Generate an SSH Key Pair on Windows

Open **Windows PowerShell** on your local computer and run the following command to create a secure `Ed25519` key pair:

```
ssh-keygen -t ed25519 -C "my_ssh_key"
```

Press `Enter` through all prompts to accept the default file location (`C:\Users\<username>\.ssh\id_ed25519`) and leave the passphrase empty for unprompted keyless logins.

2 Step 2: Transfer the Public Key to the Linux VM

Since `ssh-copy-id` is not natively available in Windows PowerShell, pipe your public key directly into the VM's `authorized_keys` file in one command:

```
Get-Content C:\Users\<username>\.ssh\id_ed25519.pub | ssh user@vm_ip "mkdir -p ~/.ssh &&
chmod 700 ~/.ssh && cat >> ~/.ssh/authorized_keys && chmod 600 ~/.ssh/authorized_keys"
```

(Enter your Linux user password one last time when prompted).

> **Alternative: Manual Copy-Paste**
>
> Run `Get-Content C:\Users\<username>\.ssh\id_ed25519.pub` , copy the output text, log into your VM terminal, and paste the line into `~/.ssh/authorized_keys` .

3 Step 3: Test the Passwordless Connection

Connect from your local Windows PowerShell to verify keyless authentication:

Method A — Standard Login (Automatic Default Key Search):

```
ssh user@vm_ip
```

Method B — Explicit Private Key Selection (`-i` flag):

```
ssh -i C:\Users\<username>\.ssh\id_ed25519 user@vm_ip
```

INFO Key Concepts & Best Practices

COMPONENT / COMMAND	LOCATION & STORAGE	PURPOSE & BEHAVIOR
Private Key `id_ed25519`	Stored on **Local Windows PC** `C:\Users\<username>\.ssh\`	Secret cryptographic identity. Must be kept secure and never shared publicly. Used by SSH to sign connection requests.
Public Key `id_ed25519.pub`	Stored on **Remote Linux VM** `~/.ssh/authorized_keys`	Acts as the lock. The local `.pub` file on Windows can even be deleted after setup without breaking login capabilities.
Standard `ssh`	Command syntax: `ssh user@vm_ip`	Automatically searches default path (`~/.ssh/`) for standard key names (`id_ed25519` , `id_rsa`).
SSH Identity (`-i`)	Command syntax: `ssh -i <path> user@vm_ip`	Explicitly specifies which private key file to use. Ideal for custom key names, AWS `.pem` keys, or shared key files.

Permissions Security Requirement

SSH will strictly ignore key files if permissions on the remote server are too loose. Ensure the remote `.ssh` directory is set to `700` (`chmod 700 ~/.ssh`) and `authorized_keys` is set to `600` (`chmod 600 ~/.ssh/authorized_keys`).

PART 2 SCP File Transfer Guide (Windows to Linux)

SCP (Secure Copy Protocol) uses your existing SSH setup to transfer files securely between local and remote machines.

1. Transfer a Single Local File to Linux VM

Send a file from your Windows PC directly to the remote user's home directory:

```
scp C:\path\to\localfile.txt user@vm_ip:~/
```

2. Transfer a File Using a Specific Private Key (`-i`)

When using a key in a custom path or non-default location, supply the `-i` flag:

```
scp -i C:\path\to\private_key C:\path\to\localfile.txt user@vm_ip:~/
```

3. Transfer an Entire Directory Recursively (`-r`)

To copy an entire folder along with all nested files and subdirectories, use the `-r` flag:

```
scp -r C:\path\to\MyFolder user@vm_ip:~/
```

Pro Tip for Windows Paths

If any folder name or file path on Windows contains spaces (e.g., `C:\My Documents\file.txt`), always wrap the path in double quotes: `"C:\My Documents\file.txt"`.